TAHA ASIM

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Portfolio | LinkedIn | GitHub

EDUCATION

Master of Management in Analytics (Part-time) | McGill University — 01/2026 – Present

Honors Bachelor of Science, Computer Science | York University — 09/2020 – 04/2025

PROFESSIONAL EXPERIENCE

Research and ML Engineer | AgenQ — 06/2026 – Present

- Researched competitor products and industry trends to evaluate how similar platforms operate and identify opportunities for improving the app.
- Translated competitive research and market analysis into a plan for upgrading the app's features and capabilities.
- Built an end to end MCP proof of concept that converts authenticated OpenAPI endpoints into AI callable tools using TypeScript, Streamable HTTP, and the official MCP SDK, giving the AI direct access to the client's backend to perform workflows automatically.
- Implemented two read only workflow tools with secure API key injection and automated integration testing, with all GitHub CI checks passing.

(Part-time) Data Scientist | DataSphere Lab — 06/2025 – Present

- Designed a data dictionary covering 45 fields across 5 datasets for a personalized restaurant discovery platform, spanning user, item, business, interaction, and QA data.
- Conducted competitive research on similar recommendation and discovery platforms to inform data architecture and modeling approach, translating findings into concrete feature engineering and algorithm decisions.
- Engineered 30+ features from user preferences and item attributes to support personalized ranking and recommendation logic.
- Prototyped recommender logic in Python using pandas, NumPy, and scikit-learn, applying feature vectors, similarity scoring, and content-based filtering.
- Wrote and documented SQL queries to extract, validate, and analyze user and interaction data, supporting data integrity and knowledge sharing across the team.

Data & Tech Intern | Toronto Hydro — 01/2023 – 12/2023

- Built an Excel and VBA driven report analyzing field electrical teams' resource usage trends, reducing retrieval travel time and cutting overall resource wastage by 17%.
- Built a 12-month operational performance dashboard and report from SAP field deployment data, improving management's visibility into field operations and contributing to a 15% reduction in resource spending.
- Partnered with project managers in an Agile environment to analyze dispatcher performance using SAP and company data, helping reduce misrouted emergency calls by ~23% and improve customer satisfaction.
- Built Python ETL pipelines to integrate structured SAP datasets, improving data reliability and supporting automated reporting workflows.

Advanced Coding Instructor | hEr Volution — 07/2022 – 09/2022

- Delivered interactive lectures to help students build programming skills and apply core concepts to real-world projects.
- Simulated debugging and QA scenarios to expose students to common development challenges and strengthen technical problem-solving.
- Conducted personalized code reviews and feedback sessions so students could learn to resolve technical issues independently.
- Facilitated structured learning and test preparation, raising average test scores by 20% by the final weeks of the course.

PROJECTS

Customer Segmentation Using Machine Learning — project link

Built a Python customer segmentation pipeline using K-Means, StandardScaler, and silhouette analysis to profile 2,216 retail customers into 3 actionable segments based on behavioral and demographic features.

ExtraaLearn Lead Conversion Scorer — project link

Developed and hyperparameter-tuned a Random Forest classifier on 4,612 EdTech leads, optimizing for recall (88%) over accuracy to minimize false negatives, with feature importance analysis identifying website engagement as the strongest conversion signal.

World Cup Match Outcome Modeling — project link

Built a football match prediction system using Python, XGBoost, and feature engineering to model outcomes from historical international results, generating win/draw/loss probabilities for World Cup fixtures.

Automated API Data Pipeline — project link

Built an automated data pipeline in Python to extract cryptocurrency market data from the CoinGecko API, transform records, and load them into PostgreSQL via SQL schema and upsert logic, with unit tests and daily GitHub Actions CI runs.

LLM-Powered Document Question Answering System — project link

Developed a Python system to query unstructured PDF documents using natural language via a RAG pipeline with OpenAI embeddings and FAISS vector search, retrieving relevant segments and generating responses grounded in source content.

AWARDS

1st Place — Experience Ventures Hackathon, York University

Collaborated with a 4-person team to design and prototype an app within 5 days connecting students seeking mental health support with peers studying or training to become therapists. Recognized for creativity, adaptability, and potential community impact.

SKILLS

Programming: Python, SQL, JavaScript, TypeScript, C++

Data & Databases: ETL Pipelines, Relational Databases, Data Mining, Snowflake

Backend & Integration: REST APIs, JSON/XML, System Integration, ETL Pipelines

Tools & Platforms: Git, GitHub, Postman, Power BI, Tableau, Excel (Pivot Tables), SharePoint, Databricks, Linux

CERTIFICATES

MIT IDSS — Data Science and Machine Learning

Anthropic — AI Fluency: Framework & Foundations